

Integration — Lesson 40 Test: BC Capstone (The Whole Toolbox)

10 mixed questions · nobody tells you the method — choose your own tool · show all work · exact answers (π , $\ln$, fractions) are fine

Name: _____ Date: _____ Score: _____ / 10

1. Evaluate $\int_1^9 (1/\sqrt{x}) \, dx$. (*all powers*)

1 pt

2. Evaluate $\int_0^2 (3x^2 - 2x) \, dx$. (*Fundamental Theorem*)

1 pt

3. Evaluate $\int_0^1 2x(x^2 + 1)^2 \, dx$. (*u-substitution*)

1 pt

4. Evaluate $\int_0^1 1/(1 + x^2) \, dx$, and give the answer exactly. (*inverse trig*)

1 pt

5. Find $\int x \cos x \, dx$, and check your answer by differentiating it. (*integration by parts*)

1 pt

6. Find $\int 1/((x - 3)(x + 1)) \, dx$. (*partial fractions*)

1 pt

7. Evaluate $\int_1^\infty 1/x^3 \, dx$, or show that it diverges. (*improper integral*)

1 pt

8. Let $g(x) = \int_2^{x^3} \sqrt{1 + t} \, dt$. Find $g'(x)$. (*FTC Part 2*)

1 pt

9. The region under $y = \sqrt{x}$ from $x = 0$ to $x = 4$ is revolved about the x -axis. Find the volume ^{1 pt} of the solid. (*volumes — disks*)

10. A particle moves along a line with velocity $v(t) = 2t - 6$ metres per second for $0 \leq t \leq 5$. ^{1 pt} Find (a) the net displacement and (b) the total distance travelled. (*net vs total*)
